BSIT-S-T-2A-T

ESP32-BASED TOMATO GREENHOUSE MONITORING SYSTEM

GROUP 4

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COMPUTER ORGANIZATION, ARCHITECTURE AND ASSEMBLY

DESCRIPTION

The ESP32-Based Tomato Greenhouse Monitoring System is an automated solution designed to enhance the efficiency and productivity of tomato cultivation in a controlled environment. This system integrates sensors, actuators, and communication modules to monitor plant growth, automate irrigation, and control artificial lighting.

The monitoring system utilizes an ESP32-CAM camera to capture images of the tomato plants, assessing their height and fruit presence. Additionally, the HC-SR04 ultrasonic sensor measures the plant’s height with accuracy. These growth parameters are compiled and sent to the user via daily text messages using the SIM800L module, providing real-time updates on the plants' condition.

DESCRIPTION

For irrigation, the system features an automated watering mechanism powered by an SG966R servo motor. This motor activates a water spray twice daily—during the morning and evening—ensuring consistent moisture levels essential for healthy plant growth.

To further enhance productivity, the system includes artificial lighting control through a relay module. During nighttime, the relay activates UV lights, providing continuous light exposure that promotes photosynthesis and supports optimal plant development.

Overall, this greenhouse monitoring system offers a cost-effective and efficient solution for tomato cultivation by reducing the need for manual intervention, ensuring consistent plant care, and providing users with real-time growth updates.

MAIN COMPONENTS AND FUNCTION

Microcontroller

- ESP32:
  - The ESP32 is the central control unit of the system.
  - It processes data from the sensors, manages communication, and controls the actuators.



ESP32

MAIN COMPONENTS AND FUNCTION

Sensors and Modules

- **ESP32-CAM:**
  - A camera module that captures images of the tomato plants.
  - It monitors the plant's physical growth by taking regular snapshots.
  - Helps the system visually assess the number of tomatoes.



ESP32-CAM

MAIN COMPONENTS AND FUNCTION

Sensors and Modules

- **HC-SR04 Ultrasonic Sensor:**
  - Measures the height of the tomato plants.
  - Uses sound waves to determine the distance between the sensor and the plant.
  - Ensures accurate plant growth monitoring.



HC-SR04

MAIN COMPONENTS AND FUNCTION

Sensors and Modules

- **PHOTORESISTOR**
  - Measures the light level of the greenhouse.
  - Uses light and changes its resistance when light is shined on it..

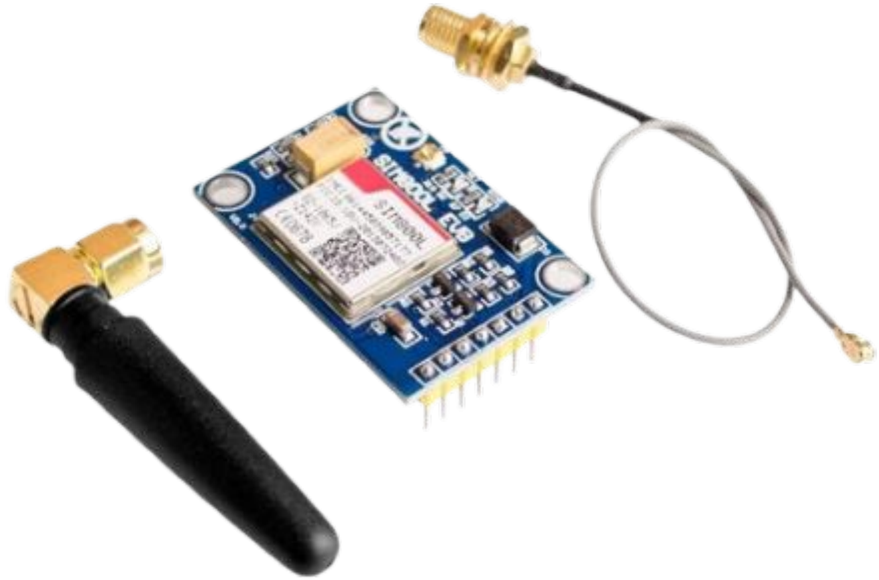


PHOTORESISTOR

MAIN COMPONENTS AND FUNCTION

Sensors and Modules

- **SIM800L Module:**
  - A GSM module used for sending SMS notifications.
  - Sends a daily text message with the plant's growth data (height, leaf count, and tomato count).
  - Provides remote updates without the need for internet access.



SIM800L

MAIN COMPONENTS AND FUNCTION

Actuators

- **SG996R Servo Motor:**
  - Controls the water spraying system.
  - Activates twice daily to irrigate the tomato plants.
  - Ensures consistent watering for healthy growth.



SG996R SERVO MOTOR

MAIN COMPONENTS AND FUNCTION

Actuators

- **RELAY**
  - Controls the UV Light.
  - Activates if the light level is low.
  - Ensures consistent light for healthy growth.



RELAY

ADDITIONAL COMPONENTS

UV Light:

- Simulates sunlight at night, supporting photosynthesis.
- Ensures tomato plants continue growing even in low-light conditions.

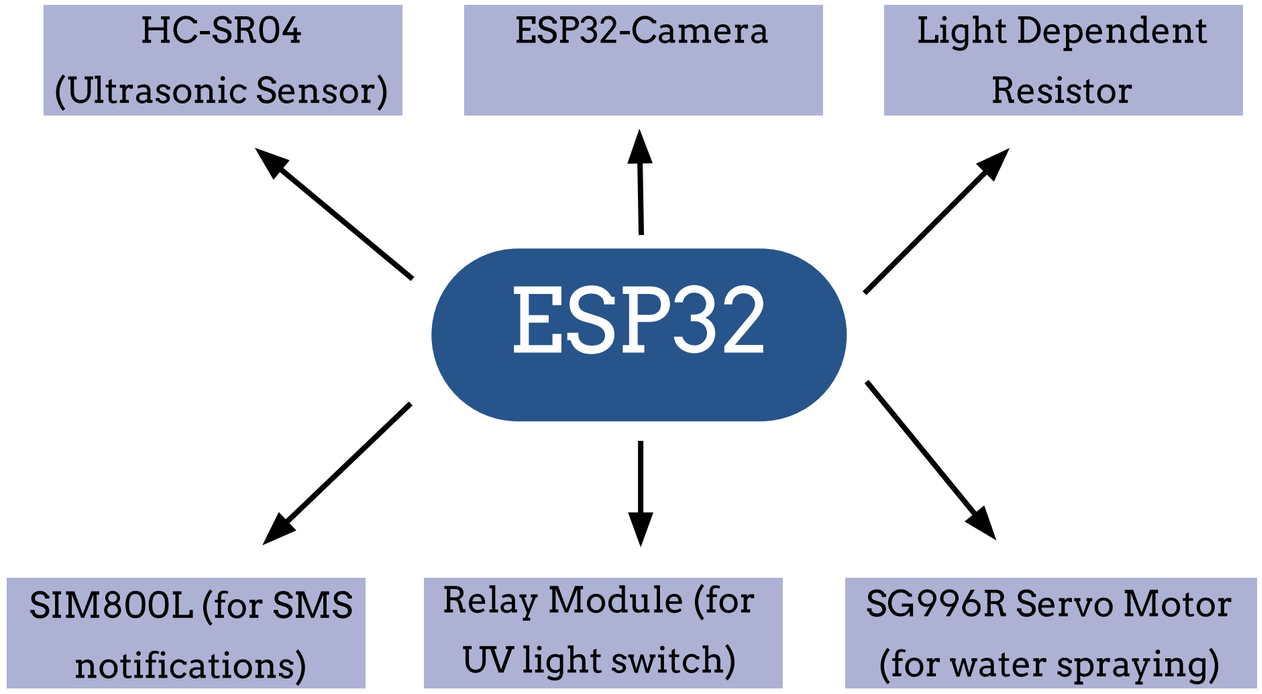


UV LIGHT

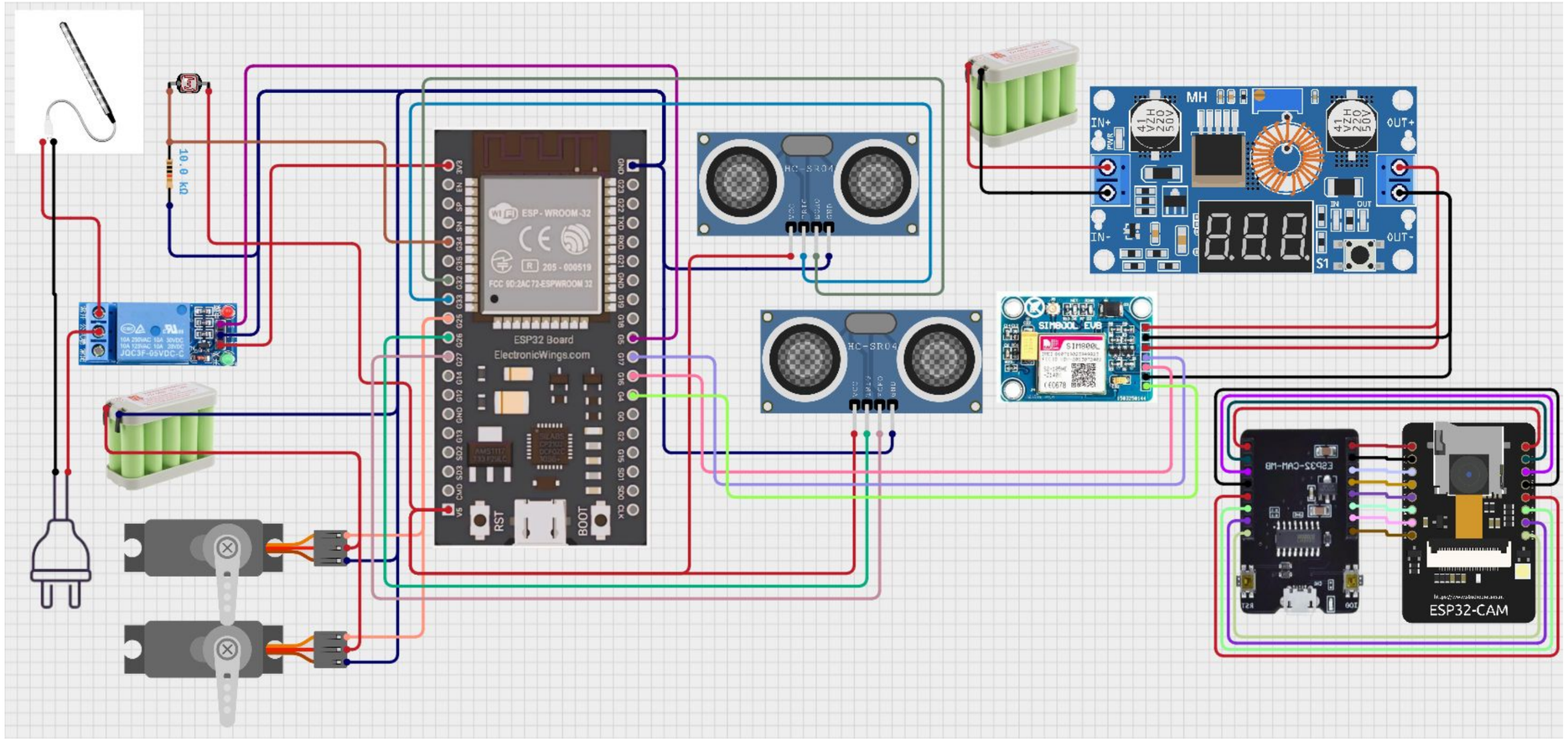
BLOCK DIAGRAM

The block diagram illustrates the system architecture, including:

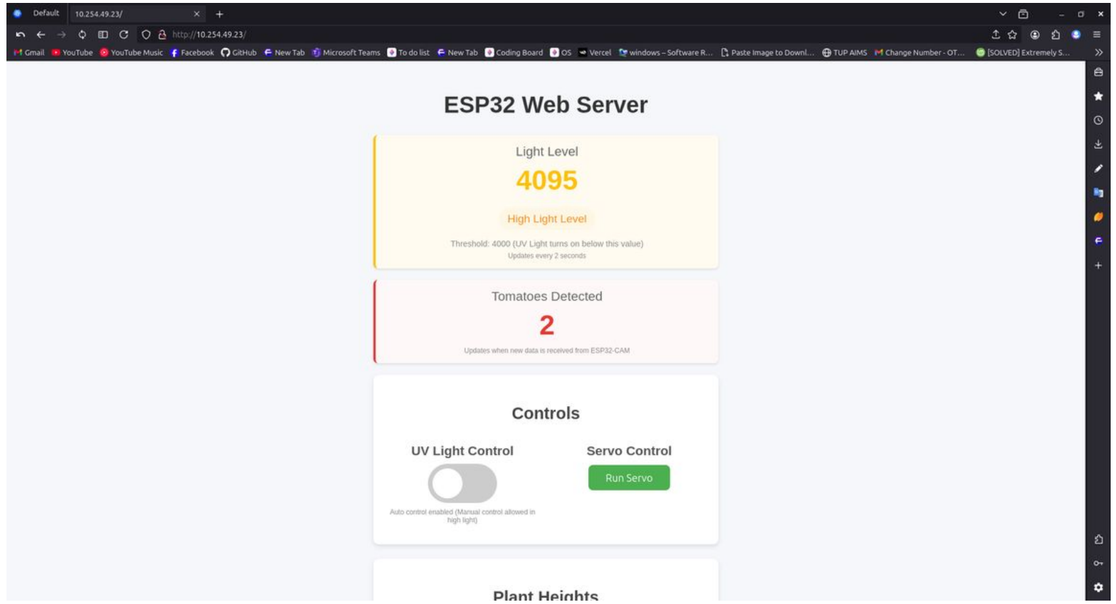
- The ESP32 microcontroller as the central unit.
- Connections to the sensors and modules (ESP32-CAM, HC-SR04, and SIM800L).
- Actuators (SG90 servo motor and relay module) for watering and light control.
- The SMS output sent to the user with the tomato plant's status.

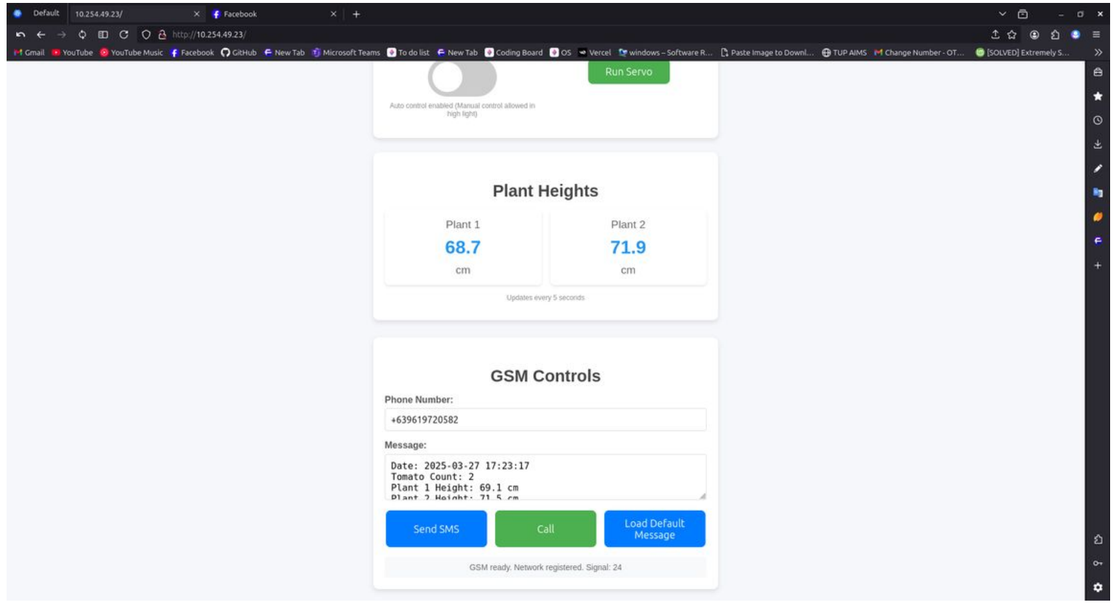


SCHEMATIC DIAGRAM



INITIAL CHECKING RESULT





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THANK YOU!